

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (IT)

November 2013

IT 202 Data Communication

Date: 30.11.2013, Saturday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Do as Direct.

- (a) A 400 – watt carrier is modulated to a depth of 75 percent. Calculate the total power in the modulated wave. [02]
 - (b) Calculate the thermal noise power available from any resistor at room temperature (290 K) for a bandwidth of 1 MHz. calculate also the corresponding noise voltage, given that $R = 50 \text{ ohm}$ [03]
 - (c) Define skin effect in pure conductor. [02]
- Q - 2 (a) Explain Tracking in super heterodyne receiver with the help of neat sketch. [08]
- (b) Give the difference between series tuned and parallel tuned circuit. [06]

OR

- Q - 2 (a) A T – type attenuator is required to provide a 6 – dB insertion loss and to match 50 ohm input and output. Find the resistor value. [06]
- (b) Write a note on sensitivity and gain of receiver. [08]
- Q - 3 (a) Write short note on Thermal Noise. [04]
- (b) Derive the relation between the output power of an AM transmitter and the depth of modulation, and plot it as a graph for values of the modulation index from zero to maximum. [05]
- (c) Write short note on RC Phase Shift oscillator. [05]

OR

- Q - 3 (a) Calculate the shot noise component of current present on a direct current of 1 mA flowing across a semiconductor junction, given that the effective noise bandwidth is 1 MHz. [04]
- (b) A certain transmitter radiates 9 kW with the carrier unmodulated, and 10.125 kW when the carrier is sinusoidally modulated. Calculate the modulation index, percent of modulation. If another sine wave, corresponding to 40 percent modulation, is transmitted simultaneously, determine the total radiated power. [05]
- (c) The tuned circuit of the oscillator in a simple AM transmitter employs a 50 – microhenry coil and a 1 – nanofarad capacitor. If the oscillator output is modulated by audio frequencies up to 10kHz, what is the frequency range occupied by the sidebands? [05]

SECTION – II

Q – 4 Do as Direct

- (a) Define double spotting in receivers. [02]
- (b) Write short note on crystal oscillator. [04]
- (c) Johnson Noise is also called as a _____. [01]

- Q – 5 (a) Draw the capacitance / reverse voltage curve for a varactor diode (voltage controlled oscillator). [02]
- (b) Give the difference between ideal filter and practical filter. [02]
 - (c) Explain mutual – inductive coupling in which the secondary voltage leads the primary current by 90 degree. [03]
 - (d) Explain diode envelope detector with necessary circuit diagram and waveforms. [03]
 - (e) Calculate the output signal to noise ratio in decibels for three identical links, given that the signal to noise ratio for any one link is 60 dB. [04]

OR

- Q – 5 (a) Describe energy spectral density with its properties. [02]
- (b) The zero bias capacitance for an abrupt junction varactor diode is 20 pF. Calculate the capacitance when a reverse bias of -7 V is applied. [02]
 - (c) Explain mutual – inductive coupling in which the secondary voltage lags the primary current by 90 degree. [03]
 - (d) Draw the block diagram of AM broadcast transmitter. [03]
 - (e) The noise figure of an amplifier is 7 dB. Calculate the output signal to noise ratio when the input signal to noise ratio is 35 dB. [04]
- Q – 6 (a) Write short note on different properties of Fourier transform. [06]
- (b) Draw the hybrid model for three different configuration of bipolar junction transistor. [08]

OR

- Q – 6 (a) What is periodic and aperiodic signal? Compare power and energy signals. [06]
- (b) Draw the generalized hybrid model for BJT and derive its following parameter. [08]
 - 1. current gain
 - 2. input impedance
 - 3. voltage gain
 - 4. output impedance
